Network Traffic Investigation Report

Incident: Suspicious External Network Communication — Reverse Shell Activity

Investigation Type: Network Traffic Analysis (PCAP)

Analysis Tool: Wireshark

Evidence File: external.pcapng

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Classification: Security Investigation

1. Executive Summary

A network traffic investigation was conducted on the packet capture file external.pcapng to examine suspicious communication involving TCP port 4444.

Analysis identified a TCP connection initiated by the Windows endpoint 192.168.132.238 to the remote endpoint 192.168.132.242. Examination of the TCP stream revealed remote command interaction, including execution of whoami and ipconfig, with command output returned by the Windows endpoint.

The communication pattern is consistent with reverse-shell activity, indicating that the Windows system was potentially under remote interactive control.

This report documents the observed network evidence, technical findings, potential security implications, and recommended defensive actions.

2. Incident Overview

The investigation focused on suspicious network communication identified in the provided packet capture.

2.1 Incident Details

Attribute	Observation
Evidence file	external.pcapng
Protocol	TCP
Source endpoint	192.168.132.238
Destination endpoint	192.168.132.242
Destination port	4444
Source system	Windows endpoint
Observed activity	Remote command execution
Assessment	Suspected reverse-shell activity

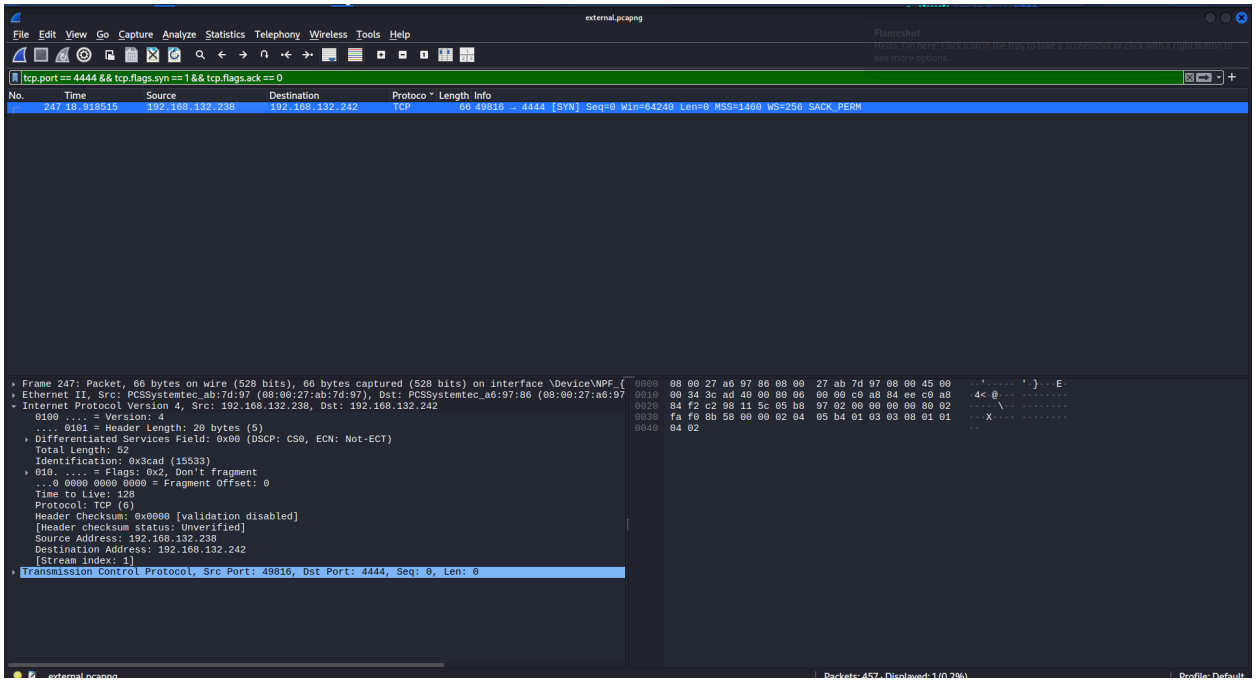


Fig: Initial TCP connection from the Windows endpoint to the remote endpoint over port 4444

2.2 Investigation Objective

The objective was to determine the nature of the communication, identify the endpoints involved, examine the exchanged data, and assess whether the traffic indicated unauthorized remote access or command execution.

3. Technical Findings

3.1 TCP Connection Establishment

The packet capture shows that the Windows endpoint (192.168.132.238) initiated a TCP connection to the remote endpoint (192.168.132.242) on destination port 4444.

Although the connection was initiated by the Windows endpoint, subsequent TCP stream analysis showed that command data was transmitted from the remote endpoint to the Windows system.

3.2 Remote Command Execution

Inspection of the TCP stream revealed command interaction involving the following Windows commands:

- whoami — used to identify the account associated with the command session.
- ipconfig — used to retrieve network configuration information.

The Windows endpoint returned command output, including the account:

desktop-tu9c8dk\emp10

The ipconfig output also identified the Windows endpoint's IPv4 address as 192.168.132.238, with subnet mask 255.255.255.0 and default gateway 192.168.132.177.



Fig: TCP stream showing remote command interaction and command output from the Windows endpoint

3.3 Communication Pattern

The observed traffic followed this pattern:

1. The Windows endpoint initiated the TCP connection to the remote endpoint.
2. The remote endpoint transmitted commands through the established connection.
3. The Windows endpoint returned command output over the same connection.

This communication pattern is consistent with reverse-shell behavior, where a system establishes an outbound connection that enables remote command interaction.

4. Impact Assessment

The observed communication presents a potential security incident involving remote command execution on a Windows endpoint.

If the activity was unauthorized, the following security implications apply:

- **Remote access:** An external endpoint appears to have interacted with a command shell on the Windows system.
- **System information exposure:** The whoami and ipconfig commands exposed account and network configuration details.
- **Further compromise risk:** An interactive command session could potentially be used to execute additional commands, access files, or establish persistence.
- **Internal network exposure:** Knowledge of the endpoint's IP configuration could assist further reconnaissance within the local network.

The packet capture confirms command interaction but does not, by itself, establish whether data was exfiltrated, persistence was established, or additional systems were compromised.

4.1 Severity Assessment

Preliminary severity: High

The apparent availability of remote command execution represents a significant security risk. This assessment is based on the observed behavior and should be validated against the system's operational context and authorization records.

5. Recommendations

The following defensive actions are recommended:

1. **Isolate the affected endpoint:** If the activity is unauthorized, restrict the Windows system's network connectivity to prevent further remote access.
2. **Investigate the remote endpoint:** Identify and validate the system at 192.168.132.242, including its ownership and purpose.
3. **Review endpoint telemetry:** Examine Windows Security logs, PowerShell logs, process creation events, and endpoint detection records for related activity.
4. **Investigate network activity:** Search firewall, DNS, and network monitoring logs for additional communication involving the identified endpoints.
5. **Check for persistence:** Examine startup entries, scheduled tasks, services, and other common persistence mechanisms.
6. **Strengthen network controls:** Restrict unnecessary outbound connections and monitor unusual traffic involving uncommon ports such as TCP/4444.
7. **Preserve evidence:** Retain the original packet capture and relevant system logs for further investigation.